DR-REMOVER: An Efficient Dynamic Object Remover Using Dual-Resolution Occupancy Grids for Constructing Static Point Cloud Maps

Guangyi Zhang, Tao Zhang, Rui Wang, Lanhua Hou

***Abstract*—Localization and navigation of vehicles using 3D point cloud maps is an important strategy in autonomous driving. In this work, we propose a new method for offline building static maps based on dual-resolution grids, called DR-REMOVER. Our method first extracts low-resolution grids containing dynamic objects, and then utilizes high-resolution grids for verifying dynamic points and reverting static points. The proposed method fully considers the sparsity of the point cloud and the environment features around the dynamic objects. As validated by experiments on SemanticKITTI and Apollo datasets, DR-REMOVER achieves more than 95% Preservation Rate (PR) for static points and Rejection Rate (RR) for dynamic points on all experimental sequences, which is significantly better than other state-of-the-art methods. Furthermore, DR-REMOVER shows excellent performance on unmanned ground vehicle (UGV) dataset with highly crowded environments. The implementation of our method is released here: https://github.com/zhongbusishaonianyou/DR-REMOVER.**

***Index Terms*—SLAM, mapping, dynamic objects.**

I. Introduction

LOCALIZATION and navigation based on 3D point cloud maps have been increasingly studied in recent years. Whether using LiDAR or vision-based sensors for relocalization relies on an accurate prior map. However, dynamic objects such as moving vehicles and pedestrians in real-world environments are inevitably scanned by LiDAR and introduced into global maps. Dynamic objects in prior maps can affect the robot's perception for environments, resulting in erroneous planning [1]. Therefore, detecting and removing dynamic objects is crucial for building high-quality static maps.

Although previously reported work utilizes clustering [2]–[4], distance images [5]–[11], and probabilistic grids [12]–[19] to identify and remove dynamic points, these works either do not have a high enough rejection rate for dynamic objects or have more false positives for static points. While building static maps offline is not limited by real-time requirements, it is still difficult to accurately distinguish between dynamic and static points on a map. On the one hand, the scan occlusion is widespread and varies with the scanning angle based on LiDAR's self-centered scan mode, and static and dynamic objects are not characterized oppositely in visibility. On the other hand, both static and dynamic objects show diverse shape features in different scans, and the sparsity of points representing the same object is not consistent. Traditional clustering methods are constrained by the sparsity of the point cloud, which makes it difficult to obtain accurate clustering results, and the scan occlusion caused by dynamic objects is difficult to be categorized and counted. In summary, the methods for removing dynamic objects still have many limitations.

In recent work, the strategy of combining grids for removing dynamic points has been widely used. These works determine the occupancy state of grids by visibility analysis or probabilistic estimation to remove or preserve points within grids.Obviously, the performance of grid-based methods is affected by the grid resolution. High-resolution grid division reduces the coexistence of the two types of points within a grid. Methods [12]–[17], [20], [21] using high-resolution grids prefer to treat each point as a basic removal unit, but grid traversal and processing inevitably consume more time. In contrast,low-resolution grids are more conducive to capturing the overall features of an object to improve the accuracy in recognizing dynamic objects, but it is difficult for such methods to differentiate between dynamic and static points within a grid. If the low-resolution grids containing dynamic objects can be accurately extracted, and then the high-resolution grids are used to differentiate between dynamic and static points within the extracted grids,the advantages of both resolution grids can potentially be combined. As a result, a new method for removing dynamic points, DR-REMOVER, is proposed following the above idea.

Therefore, how to extract low-resolution grid regions containing dynamic objects becomes critical. In fact, the regions that the dynamic object passes through have a significant change in occupancy height when a dynamic object appears on a roadway. The previous works [22]–[24] have proved the feasibility of utilizing a height feature to extract dynamic regions. In this work, we consider the effects of line-of-sight occlusion, grid resolution and dynamic point distribution rather than just utilizing height features to extract dynamic regions. The combination of dual-resolution occupancy grids is also used to motivate the proposed method to exhibit excellent performance. The contributions of the proposed method are as follows:

1) A new method for building static maps is proposed based on dual-resolution grids and combined with height features for removing dynamic objects such as vehicles and pedestrians.

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2) We combine the Occupancy Height Ratio(OHR), Full Occlusion Distance(FOD), and Area Occupancy Ratio(AOR) to improve the accuracy of dynamic region detection.
3) Two-step reversion of the false positives to ground points utilizing Cell classification and spatial distribution test.
4) Extensive evaluation on multiple datasets. Quantitative comparison with state-of-the-art methods on SemanticKITTI [25] and Apollo dataset [26]. Furthermore, the robustness of the proposed method is validated on UGV dataset with a highly crowded environment.

II. Related Work

Obviously, the motion state of an object cannot be analyzed using a single frame of LiDAR scan data. Whether removing dynamic points online using neighboring multi-frame scans or removing dynamic points offline based on a map, they are both essentially dedicated to opposing the features of dynamic and static points for reducing the overlap in features between the two types of points. The previously reported work mainly utilizes the feature differences between dynamic and static points in the following three aspects: (1) motion state relative to the carrier, (2) scan changes from line-of-sight occlusion, (3) height changes within local regions. According to the features used, these works are categorized into: segmentation-based methods, scan occlusion-based methods, and occupancy height-based methods.

Segmentation-based methods. Estimation of object motion states relies on point cloud segmentation. Earlier works used clustering to segment the point cloud and then estimated the object's motion by calculating the change in position of the object's center of mass. Compared with traditional clustering methods, deep learning-based semantic segmentation methods have become more popular in recent years due to their excellent segmentation accuracy. Unfortunately, purely semantic segmentation methods [27]–[31] do not have the ability to estimate the motion state of an object, such as they cannot distinguish whether a vehicle is moving or stationary. As a result, some methods [6], [10], [32]–[34] of deep learning for LiDAR moving object segmentation (MOS) have been proposed. LMNet [32] implemented online motion object segmentation using distance images as input to a convolutional neural network and combining residual images. RVMOS [6] performed feature extraction based on distance images and used data enhancement methods of time interval modulation and zero residual image synthesis to improve segmentation performance. 4DMOS [33] developed a receding horizon strategy that not only predicts moving objects online, but also improves the prediction at any time based on new observations. Auto-MOS [35] automatically generates point cloud labels to improve the performance of existing learning-based MOS systems. Nonetheless, deep learning-based methods cannot directly output static point cloud maps.

Scan occlusion-based methods. The appearance of dynamic objects in closer positions causes LiDAR scanning to be terminated earlier. Both ray-tracing-based and visibility-based methods utilize scanning differences resulting from line-of-sight occlusion. Ray-tracing-based methods [13], [15], [16], [20], [36] require high-resolution voxel segmentation and use an occupancy probability strategy to update the occupancy state of all voxels. Octomap [16] proposed an efficient and compact map representation based on octree and probabilistic occupancy estimation. With Octomap's representation, Octomap-KNN [15] introduced a ground segmentation strategy to update the occupancy state of only the voxels where non-ground points are located, and accelerated occupancy map building by using object detection for object classification. Peopleremover [20] used a fast voxel traversal algorithm and a spherical quadtree structure to reduce computational cost. Overall, ray-tracing-based methods treat all voxels as equally important, and it is more costly to maintain the map and update the occupancy state of all voxels.

In contrast, visibility-based methods [5], [9], [11], [37]–[39] focus more on scanning changes due to occlusion and are not concerned with the occupancy state of voxels through which the scanning rays pass. Wen et al. [38] proposed a 3D spatial model based on scanning rays and localized point distributions, and interpolated the occupancy between rays by treating the ray shapes as triangular prisms. Removert [9] used multi-resolution distance images for reverting static points, reducing the effects of LiDAR motion or alignment errors. Unlike Removert 's reverting to static points, DynamicFilter [37] introduced the map-based reverting to minimize false detections caused by large incidence angles and occlusions. Park et al. [5] proposed a probability-based nonparametric background estimation method to address the limitations of visibility methods by accumulating differential distance images. DORF [11] proposed a backward horizon sampling mechanism to retain the identified static points and used polarized elevation maps to recover the ambiguous static points.

Scan occlusion-based methods should aim to avoid false predictions caused by the occlusion of static objects. Unfortunately, scan occlusion is affected by the road environment, the scanning angle of rays, the motion of carriers, and many other factors, and the complexity and variability of occlusion scenarios limit the performance of such methods.

Occupancy height-based methods. Dynamic objects do not have the fixity of space occupied by static objects, making the region through which dynamic objects pass exhibit obviously different occupied heights in different LiDAR scans. Asvadi et al. [22] constructed a 2.5D grid for motion detection and estimated static ground regions using height-mean and height-variance. Kim et al. [24] built grid maps of real environments based on multi-layered vertical information for real-time maintenance and updating. ERASOR [23] took the same plane division method as Scan Context [40] and utilized height differences for scan ratio tests to detect dynamic regions.

The letter presents a new method for offline building of static maps for SLAM post-processing, inspired by ERASOR [23] and Removert [9]. Unlike previous grid-based methods, DR-REMOVER uses different processing units for dynamic object detection and removal, that is, the low-resolution grid is used as the dynamic object detection unit and the high-resolution grid is used as the dynamic object removal unit. The separation of the detection and removal units allows the

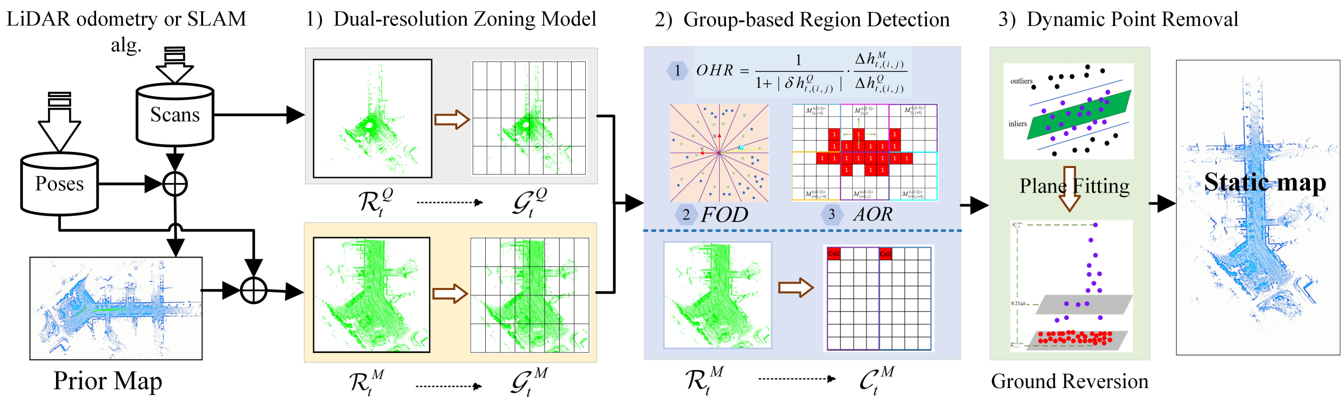


Fig. 1. Algorithm overview. DR-REMOVER consists of three main sections: dual-resolution zoning model, Group-based region detection and dynamic point removal. Firstly, extract the valid points $\mathcal{R}_t^Q$ of the current scanned data $\mathcal{P}_t^Q$ and the valid points $\mathcal{R}_t^M$ of the corresponding submap $\mathcal{P}_t^M$, and then the Group-based zoning models $\mathcal{G}_t^M$ and $\mathcal{G}_t^Q$ are created for $\mathcal{R}_t^M$ and $\mathcal{R}_t^Q$,respectively. We additionally create a Cell-based zoning model $\mathcal{C}_t^M$ for $\mathcal{R}_t^M$. Next,Groups containing dynamic points are jointly detected using OHR, FOD and AOR factors. Finally,dynamic points in the raw map are removed by region ground fitting and ground point reversion techniques.

advantages of both resolution grids to be fully utilized. The proposed method focuses on removing objects that change during a SLAM period and cannot be used for long-term change detection.

III. Methodology

In this section, the removal-revert mechanism based on dual-resolution grids is described specifically. First, we introduce a dual-resolution zoning model. Next, we describe the process of using dual-resolution grids to jointly detect dynamic regions. At the end, we explain the steps for determining dynamic points using ground plane fitting and ground point reversion. The overall pipeline of our method is depicted in Fig. 1.

A. Problem Definition

The SLAM algorithm can accomplish the estimation of the LiDAR poses by point cloud registration and output the LiDAR odometry or optimized poses.These corrected or optimized LiDAR poses are often used to merge LiDAR scans to build 3D point cloud maps. Let $\mathcal{P}_t^Q$ be the scanned frame at time t, ${}_Q^W\mathbf{T}_t$ is the transformation matrix of the LiDAR frame Q at time t with respect to the world frame W. The prior map $\mathcal{M}$ in the world frame can be built as follows:

$$\mathcal{M} = \bigcup_{t\in\Gamma} {}_Q^W\mathbf{T}_t \boxplus \mathcal{P}_t^Q, \tag{1}$$

where Γ denotes the set of scan data timestamps, $\boxplus$ represents the homogeneous transform operation between the LiDAR frame Q and the world frame W. Clearly, the prior map $\mathcal{M}$ as a set of scan data contains all measured dynamic points.

We extract the submap associated with $\mathcal{P}_t^Q$ coordinates from $\mathcal{M}$ and obtain the submap $\mathcal{P}_t^M$ in the LiDAR frame by ${}_W^Q\mathbf{T}_t = {}_Q^W\mathbf{T}_t{}^{-1}$. In this letter, superscripts Q and M are used to denote the LiDAR scan frame and the submap, respectively.The proposed method aims to identify and remove dynamic points in $\mathcal{P}_t^M$ by comparing the feature differences between $\mathcal{P}_t^Q$ and $\mathcal{P}_t^M$, and ultimately obtain a global static map:

$$\hat{\mathcal{M}} = \mathcal{M} - \bigcup_{t\in\Gamma} \mathcal{M}_{t,dyn}, \tag{2}$$

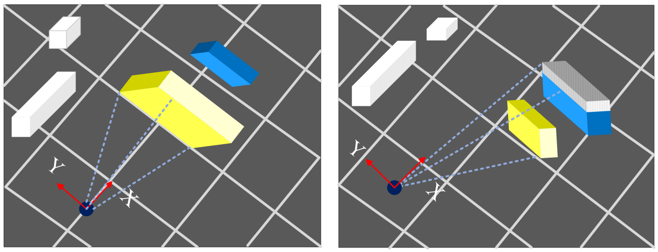
where $\mathcal{M}_{t,dyn}$ denotes dynamic points identified from $\mathcal{P}_t^M$ by one comparison of $\mathcal{P}_t^Q$ and $\mathcal{P}_t^M$.

B. Dual-resolution Zoning Model

The proposed method compares the feature differences between $\mathcal{P}_t^Q$ and $\mathcal{P}_t^M$ within the same subregion when performing dynamic object removal.This region-based feature comparison strategy requires regional division and saving points and feature information separately for each subregion. Before performing the zoning, we redefine the effective area and take $\mathcal{R}_t = \{\mathbf{p}_k | |x_k| < l, |y_k| < l, h_{\min} < z_k < h_{\max}\}$ as the set of effective points, where (x_k, y_k, z_k) are the coordinates of the point $\mathbf{p}_k$ in the LiDAR frame, $h_{\min}$ and $h_{\max}$ are the heights relative to the ground, and l is the size parameter of an effective plane area. Considering the height of dynamic objects such as moving vehicles and pedestrians on the road, we let $h_{\max} = 3$ m, $h_{\min} = -1$ m and set $l = 60$ m.

Group-based Zoning Model In the previously reported work, the plane zoning is based on two modes: the first is the division along the x-axis and y-axis of the LiDAR frame, obtaining grids of the same size. The second is the division along the radial and annular directions centered on the origin of the LiDAR frame to obtain sector rings of unequal size. The second method focuses on compensating for the sparsity of the point cloud, whereas we are more concerned with the completeness of the dynamic object features within each subregion. The low-resolution grid obtained by zoning along the direction of the coordinate axes is more prone to contain the majority of the scan points of a dynamic object.

Thus, the effective area of $l \times l$ m^2 in the xy-plane is divided into $N_x \times N_y$ equal-sized grids,where N_x and N_y are the number of grids divided along the x-axis and y-axis, respectively. Importantly, the grids resulting from the division



(a) Fully occluded scene (b) Partially occluded scene

Fig. 2. Two scenarios of scan occlusion. The yellow-filled rectangles represent the objects causing scan occlusion, and the blue-filled rectangles represent the occluded objects or parts of the occluded objects.

have the larger size of $4\text{ m} \times 2\text{ m}$, i.e., set $N_x = 60$, $N_y = 30$. This low-resolution rectangular grid is called a Group, which is similar to a tissue in biology, indicating that it is composed of multiple similar cells. The points in $\mathcal{R}_t$ fall into multiple Groups after the effective area is divided, and the set of points in each Group can be defined as:

$$\begin{aligned}\mathcal{G}_{t,(i,j)} = \{\mathbf{p}_k | &\frac{(i-1)\cdot l}{N_x} - l \le x_k < \frac{i \cdot l}{N_x} - l, \\ &\frac{(j-1)\cdot l}{N_y} - l \le y_k < \frac{j \cdot l}{N_y} - l\},\end{aligned} \tag{3}$$

where (i, j) is the index of the Group. Denote the Group-based zoning model as $\mathcal{G}_t$, which can be defined as:

$$\mathcal{G}_t = \bigcup_{i\in[N_x],j\in[N_y]} \mathcal{G}_{t,(i,j)}. \tag{4}$$

We create Group-based zoning models for sub-maps and scan frames, denoted as $\mathcal{G}_t^Q$ and $\mathcal{G}_t^M$, respectively.

Cell-based Zoning Model As described in related work, the coexistence of static and dynamic points within a low-resolution grid tends to cause serious false detections of static points, and thus it is undesirable to use Group as a whole region for dynamic point removal. A better strategy is to divide the effective area $\mathcal{R}_t^M$ of the submap again, using high-resolution grids to remove the dynamic points contained in low-resolution grids. Using the same zoning pattern, the effective region is divided into $N_c \times N_c$ high-resolution squares, where N_c is the number of grids divided along the direction of the two coordinate axes. We set $N_c = 240$ to obtain the $0.5\text{ m} \times 0.5\text{ m}$ higher resolution grid, which is called Cell, meaning indivisible basic functional unit.

The points in $\mathcal{R}_t^M$ fall into multiple Cells after the effective area is divided, and the set of points in each Cell can be defined as:

$$\begin{aligned}\mathcal{C}_{t,(m,n)}^M = \{\mathbf{p}_k \in \mathcal{R}_t^M | &\frac{(m-1)\cdot l}{N_c} - l \le x_k < \frac{m \cdot l}{N_c} - l, \\ &\frac{(n-1)\cdot l}{N_c} - l \le y_k < \frac{n \cdot l}{N_c} - l\},\end{aligned} \tag{5}$$

where (m, n) is the index of the Cell, Denote the Cell-based zoning model as $\mathcal{C}_t^M$, which can be defined as:

$$\mathcal{C}_t^M = \bigcup_{m\in[N_c],n\in[N_c]} \mathcal{C}_{t,(m,n)}^M \tag{6}$$

C. Group-based Region Detection

The dynamic points in $\mathcal{P}_t^M$ are divided into multiple discrete Groups after the Group-based zoning model is created. We use the Groups as the basic detection unit and label all Groups where dynamic points exist. In this section, Groups with dynamic points are jointly determined using Occupancy Height Ratio (OHR), Full Occlusion Distance(FOD) and Area Occupancy Ratio (AOR). For simplified notation, we use Ah and Ih to denote the maximum and minimum heights, respectively. And we use the index symbols of the set of points as subscripts for Ah and Ih to select the set of points to be processed.

Occupancy Height Ratio Height changes in a Group caused by a dynamic object are significantly different from the undulations of the ground. This distinctive feature is used to define OHR to quantify the difference between $\mathcal{G}_{t,(i,j)}^Q$ and $\mathcal{G}_{t,(i,j)}^M$ in occupancy height. The OHR is defined as follows:

$$OHR = \frac{1}{1 + |\delta h_{t,(i,j)}|} \cdot \frac{\Delta h_{t,(i,j)}^M}{\Delta h_{t,(i,j)}^Q}, \tag{7}$$

where $\Delta h_{t,(i,j)}^M = Ah_{t,(i,j)}^M - Ih_{t,(i,j)}^M$,$\Delta h_{t,(i,j)}^Q = Ah_{t,(i,j)}^Q - Ih_{t,(i,j)}^Q$ and $\delta h_{t,(i,j)} = Ih_{t,(i,j)}^Q - Ih_{t,(i,j)}^M$.Our method introduces a reconciliation coefficient $1/(1 + |\delta h_{t,(i,j)}|)$ compared with the Scan Ratio Test (SRT) in ERASOR [23]. This coefficient is used to avoid obvious height changes caused by static objects: One scenario is the ground position shift caused by inaccurate mapping, and the other scenario is the restricted scan of partially occluded objects, as depicted in Fig. 2(b).

Without considering LiDAR scan occlusion, a static object has approximately the same occupancy height in Group of $\mathcal{G}_t^M$ and $\mathcal{G}_t^Q$, making the value of OHR close to 1. However, the value of OHR increases significantly when the dynamic object only occupies in the Group of $\mathcal{G}_t^M$. Therefore, a Group is considered to contain dynamic points when $OHR > thres$, and this Group is labeled as dynamic. $thres$ is the experimental setting parameter.

Full Occlusion Distance At some moments, some static objects are hidden behind higher objects and the LiDAR cannot scan them. Although $\mathcal{G}_{t,(i,j)}^Q$ does not contain points of static objects that are fully occluded, these points appear in $\mathcal{G}_{t,(i,j)}^M$. Obviously, the (i, j)-th Group of $\mathcal{G}_t^M$ is incorrectly labeled as dynamic using Eq. (7). Therefore, full scan occlusion presented from Fig. 2(a) in LiDAR scans needs to be ruled out.

The xy-plane area of $\mathcal{R}_t^M$ is divided in equal angles according to the rotational scanning mode of LiDAR. Let $\theta = \operatorname{atan2}(y_k, x_k)$, the points in each sub-region can be represented as:

$$\mathcal{J}_{t,s}^M = \{\mathbf{p}_k \in \mathcal{R}_t^M | \frac{(\mathrm{s}-1)\cdot 2\pi}{N_s} \le \theta < \frac{\mathrm{s} \cdot 2\pi}{N_s}, z_k > \kappa\}, \tag{8}$$

where N_s is the number of sub-regions. s is the index of sub-regions, κ is the height threshold. We use the minimum distance value $d_{t,s} = \min\{d_k | d_k = \sqrt{{x_k}^2 + {y_k}^2}, (x_k, y_k) \in \mathbf{p}_k, \mathbf{p}_k \in \mathcal{J}_{t,s}^M\}$ to assign a value for each sub-region. A sub-region is assigned the value l if the set of points in the sub-

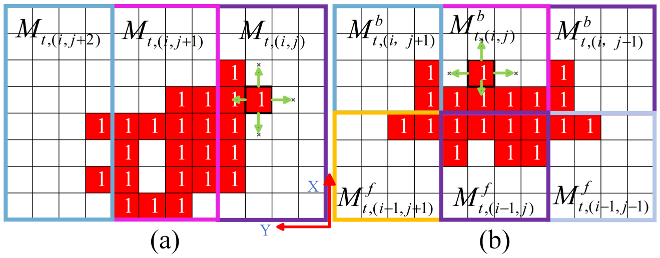


Fig. 3. Definition of Area Occupancy Ratio. Each small grid represents a Cell, and the bold colored boxes are the state matrices of the different Groups. The red filled grids are the occupied cells searched by BFS, and the smaller green arrows indicate the four directions of the search. (a) The occupied cells are concentrated in the left half of the (i, j)-th Group. (b) The occupied cells are concentrated in the back half of the (i, j)-th Group.

region is empty. The occlusion state of each Group is estimated using FOD, which is defined as follows:

$$FOD = \sqrt{{x^M_{t,(i,j)}}^2 + {y^M_{t,(i,j)}}^2} - d_{t,s} - \frac{2l}{N_x}, \tag{9}$$

where $(x^M_{t,(i,j)}, y^M_{t,(i,j)})$ denote the coordinates of the maximum height point within the (i, j)-th Group. The value of s is the index of the sub-region where $(x^M_{t,(i,j)}, y^M_{t,(i,j)})$ is located. The current Group is considered to be fully occluded when $FOD > 0$.

Area Occupancy Ratio The points in $\mathcal{P}^Q_t$ usually only describe the partial contours of some objects due to the limitation of the LiDAR scan angle at time t. If the points in $\mathcal{P}^M_t$ can describe the whole contour of a static object and the unscanned portion of that static object is individually divided into a Group, this Group is also incorrectly labeled as dynamic using Eq. (7). Therefore, we estimate the size and distribution of the Cell occupied by an object to avoid such situation.

First, estimate the Cells occupied by objects within the dynamic Groups. We use the height information to determine the Cell's occupancy state and define I_1 as:

$$I_1 = \frac{Ah^M_{t,(i,j)} - Ah^M_{t,(m,n)}}{Ah^M_{t,(i,j)}}. \tag{10}$$

Let $N_{cx} = N_c/N_x, N_{cy} = N_c/N_y$, all Cells in the (i, j)-th Group whose indexes satisfy $m \in [N_{cx} \cdot i, N_{cx} \cdot (i + 1))$, $n \in [N_{cy} \cdot j, N_{cy} \cdot (j+1))$. A Cell is considered to be occupied by an object when $I_1 < 0.5$. We use the binary state matrix $\mathbf{M}_{t,(i,j)}$ to reflect the occupancy state of all Cells within the (i, j)-th Group at time t.

Then, merge neighboring state matrices to search for all Cells occupied by an object. The state matrix $\mathbf{M}_{t,(i,j)}$ is divided into left, right, front, and back halves, and the directions of the halves are denoted using l , r , f and b, respectively. We define the factor I_2 to estimate the centralized orientation of occupied Cells in $\mathbf{M}_{t,(i,j)}$,and I_2 is defined as:

$$I_2 = \frac{|N^l_{t,(i,j)} - N^r_{t,(i,j)}|}{|N^f_{t,(i,j)} - N^b_{t,(i,j)}|}, \tag{11}$$

where $N^l_{t,(i,j)}$, $N^r_{t,(i,j)}$,$N^f_{t,(i,j)}$, $N^b_{t,(i,j)}$ are the number of Cells occupied by the four halves of $\mathbf{M}_{t,(i,j)}$, respectively. We consider that the object mainly occupies the left or right half of the (i, j)-th Group when $I_2 > 1$. In this case, the rules for merging neighboring Groups are defined as follows:

$$\begin{cases} \mathbf{M}_{t,(i,j-2)} \oplus \mathbf{M}_{t,(i,j-1)} \oplus \mathbf{M}_{t,(i,j)} & \text{if } N^l_{t,(i,j)} > N^r_{t,(i,j)} \\ \mathbf{M}_{t,(i,j)} \oplus \mathbf{M}_{t,(i,j+1)} \oplus \mathbf{M}_{t,(i,j+2)} & \text{else} \end{cases}, \tag{12}$$

where $\oplus$ is defined as the horizontal splicing of matrices of the same size. We consider that the object mainly occupies the upper or lower half of the Group when $I_2 \leq 1$. In this case, the rules for merging neighboring Groups are defined as:

$$\begin{cases} (\mathbf{M}^b_{t,(i,j-1)} \oplus \mathbf{M}^b_{t,(i,j)} \oplus \mathbf{M}^b_{t,(i,j+1)}) & \text{if } N^f_{t,(i,j)} \leq N^b_{t,(i,j)} \\ \otimes (\mathbf{M}^f_{t,(i+1,j-1)} \oplus \mathbf{M}^f_{t,(i+1,j)} \oplus \mathbf{M}^f_{t,(i+1,j+1)}) & \\ (\mathbf{M}^b_{t,(i-1,j-1)} \oplus \mathbf{M}^b_{t,(i-1,j)} \oplus \mathbf{M}^b_{t,(i-1,j+1)}) & \text{else} \\ \otimes (\mathbf{M}^f_{t,(i,j-1)} \oplus \mathbf{M}^f_{t,(i,j)} \oplus \mathbf{M}^f_{t,(i,j+1)}) & \end{cases}, \tag{13}$$

where $\otimes$ is defined as the same size matrix for vertical splicing.

In the final step, the Breadth-First Search (BFS) is used to find connected regions in the merged state matrices. The search process using BFS is depicted as Fig. 3. The BFS starts searching only from the Cell that contains the maximum height point as this Cell is occupied by an object. We use the number of Cells contained in a connected region to reflect the occupancy size of an object. Introduce AOR as:

$$AOR = \frac{N^{in}_{t,(i,j)}}{N^{tot}_{t,(i,j)}}, \tag{14}$$

where $N^{in}_{t,(i,j)}$ is the number of connected Cells within the (i, j)-th Group and $N^{tot}_{t,(i,j)}$ is the total number of connected Cells. The dynamic Group labeled using Eq. (7) reverts to static when $AOR \leq 0.5$. In other words, the dynamic Group should contain the majority of the region occupied by a dynamic object.

In summary, a Group is considered as containing dynamic points that need to satisfy: (1) $OHR > thres$, (2) $FOD \leq 0$, and (3) $AOR > 0.5$.

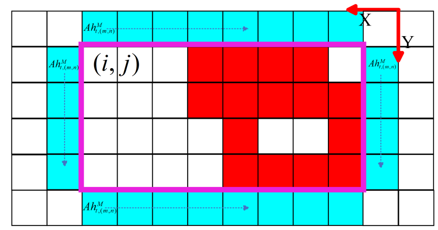
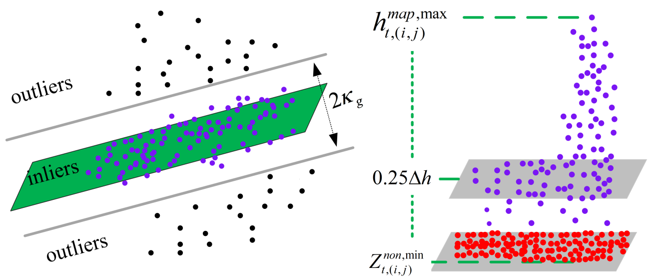


Fig. 4. Rough estimation of ground level.The red-filled Cells contain dynamic points, and the $Ah^M_{t,(m,n)}$ of blue-filled Cells are used to estimate ground height.

D. Cell-based Dynamic Object Removal

In previous region-based dynamic point removal methods, a prevalent strategy is to use all the points in the dynamic region for ground plane fitting and remove non-ground points as dynamic points. This implies that each dynamic area is regarded as an indivisible unit for dynamic point removal.



(a) Ground plane fitting (b) Spatial distribution test

Fig. 5. Ground plane fitting process and spatial distribution test. (a) The green plane is the fitted ground plane, and the purple points represent the new seed points selected based on the ground plane equation and edge threshold. (b) The red points represent the ground points that were successfully clustered, and the purple points are the final removed dynamic points within the Group.

Unfortunately, this strategy can mistakenly remove large areas of ground points because of inaccurate ground fitting. Consequently, we created Cells to separate the dynamic Groups and used Cells as dynamic point removal units to minimize the false positives of static points.

Cell Classification Ground areas usually exist around dynamic objects, and we look for static Cells adjacent to labeled Groups to estimate the ground level. As illustrated in Fig. 4, all Cells adjacent to the Group are marked in blue. Let $\mathcal{H}^M_{t,(i,j)}$ be the $Ah^M_{t,(m,n)}$ set of blue marked Cells and take $Gh^M_{t,(i,j)} = \min\{\mathcal{H}^M_{t,(i,j)}\}$ as the ground level value. The Cell with index (m,n) is labeled as static if $Ah^M_{t,(m,n)} < Gh^M_{t,(i,j)}$. In particular, the points within the static Cell are treated as static points and are not used in the next step of ground plane fitting. This processing preserves all points in the static Cell, although it uses fewer points for ground plane fitting.

Ground-plane Fitting Let the set of points within the dynamic Cell in the (i,j)-th Group be $\mathcal{S}^M_{t,(i,j)}$, and use $Gh^M_{t,(i,j)}$ to perform point cloud filtering on $\mathcal{S}^M_{t,(i,j)}$ to eliminate wild values. The filtered points satisfy:

$$\mathcal{F}^M_{t,(i,j)} = \{\mathbf{p}_k \in \mathcal{S}^M_{t,(i,j)} | z_k > Gh^M_{t,(i,j)} - \kappa_g\}, \tag{15}$$

where κ_g is the edge threshold. The initial seed points are extracted from $\mathcal{F}^M_{t,(i,j)}$, and the resulting seed points set is represented as follows:

$$^0\ell_{t,(i,j)} = \{\mathbf{p}_k \in \mathcal{F}^M_{t,(i,j)} | z_k < \bar{z} + \mathrm{h}_{seed}\}, \tag{16}$$

where $\bar{z}$ is the average height of seed points and h_{seed} is the height threshold. The covariance matrix $\mathbf{L}_{t,(i,j)}$ of seed points is calculated as:

$$\mathbf{L}_{t,(i,j)} = \sum_{k=1}^{N_{seed}} (\mathbf{p}_k - \bar{\mathbf{p}}_{t,(i,j)})(\mathbf{p}_k - \bar{\mathbf{p}}_{t,(i,j)})^T, \tag{17}$$

where N_{seed} is the number of seed points and $\bar{\mathbf{p}}_{t,(i,j)}$ is the average position of all seed points. Next, Principal Component Analysis (PCA) is used to solve for the three eigenvalues of $\mathbf{L}_{t,(i,j)}$ and the corresponding three eigenvectors. We take the eigenvector $\mathbf{n}_{t,(i,j)} = [a_{t,(i,j)}, b_{t,(i,j)}, c_{t,(i,j)}]^T$ corresponding to the smallest eigenvalue as the normal vector of the ground, then the fitted plane equation is expressed as follows:

$$a_{t,(i,j)}x + b_{t,(i,j)}y + c_{t,(i,j)}z + d_{t,(i,j)} = 0, \tag{18}$$

where the coefficient $d_{t,(i,j)} = -\mathbf{n}^T_{t,(i,j)}\bar{\mathbf{p}}_{t,(i,j)}$. Using the plane equation of Eq.(24), the set of new seed points is obtained as:

$$^1\ell_{t,(i,j)} = \{\mathbf{p}_k \in \mathcal{F}^M_{t,(i,j)} | d_{t,(i,j)} - \hat{d}_k < \kappa_g\}, \tag{19}$$

where $\hat{d}_k = -\mathbf{n}^T_{t,(i,j)}\mathbf{p}_k$. Fig. 5(a) illustrates the process of selecting the interior points as seed points based on the fitted plane equations.

The process of Eqs. (16-19) is repeated for three iterations and the seed points generated from the last iteration are used as ground points. In particular, if fitting the ground plane equation fails, our method treats all points in $\mathcal{F}^M_{t,(i,j)}$ as non-ground points for the next steps. Such an operation enables our method to recognize dynamic objects without ground points.

E. Reverting Ground Points

Elevation Filtering The points of static Cells are used to estimate the ground height within each dynamic Group, compensating for the false positives of ground points caused by inaccurate or failed ground fitting. We perform a single reversion to ground points using the mean height $\bar{h}^M_{t,(i,j)}$, which is computed as:

$$\bar{h}^M_{t,(i,j)} = \frac{1}{N_{st}} \sum_{k=1}^{N_{st}} Ah^M_{t,(m,n)}, \tag{20}$$

where N_{st} denotes the number of static Cells in the (i,j)-th Group. The points with heights below $\bar{h}^M_{t,(i,j)}$ are reverted to ground points.

Spatial Distribution Test The entire process of spatial distribution testing is displayed in Fig. 5(b). Let the set of non-ground points be $\mathcal{X}^M_{t,(i,j)}$, we divide the height range of points in $\mathcal{X}^M_{t,(i,j)}$ into four intervals and count only the points within the lowest height interval. The set of points $\mathcal{V}^M_{t,(i,j)}$ in the lowest interval is denoted as:

$$\mathcal{V}^M_{t,(i,j)} = \{\mathbf{p}_k \in \mathcal{X}^M_{t,(i,j)} | z_k < \frac{Ah^M_{t,(i,j)} + 3h^M_{t,(i,j)}}{4}\}, \tag{21}$$

where $h^M_{t,(i,j)}$ is the minimum height of points in $\mathcal{X}^M_{t,(i,j)}$. Then, the number of points $N^{low}_{t,(i,j)}$ in $\mathcal{X}^M_{t,(i,j)}$ and the number of points $N^{non}_{t,(i,j)}$ in $\mathcal{V}^M_{t,(i,j)}$ are computed, the quantity ratio is used to reflect the spatial distribution offset. The Vertical Offset Factor (VOF) is defined as:

$$VOF = \frac{N^{low}_{t,(i,j)}}{N^{non}_{t,(i,j)}}. \tag{22}$$

The points of the dynamic object are more uniformly distributed vertically making the value of VOF approximate 0.25. Therefore, the set of non-ground points $\mathcal{X}^M_{t,(i,j)}$ is considered to still contain ground points when $VOF > 0.5$. We perform Euclidean clustering for $\mathcal{V}^M_{t,(i,j)}$ using KD-Tree. The points in the largest class are reverted to static points, and the other points are removed from $\mathcal{P}^M_t$ as identified dynamic

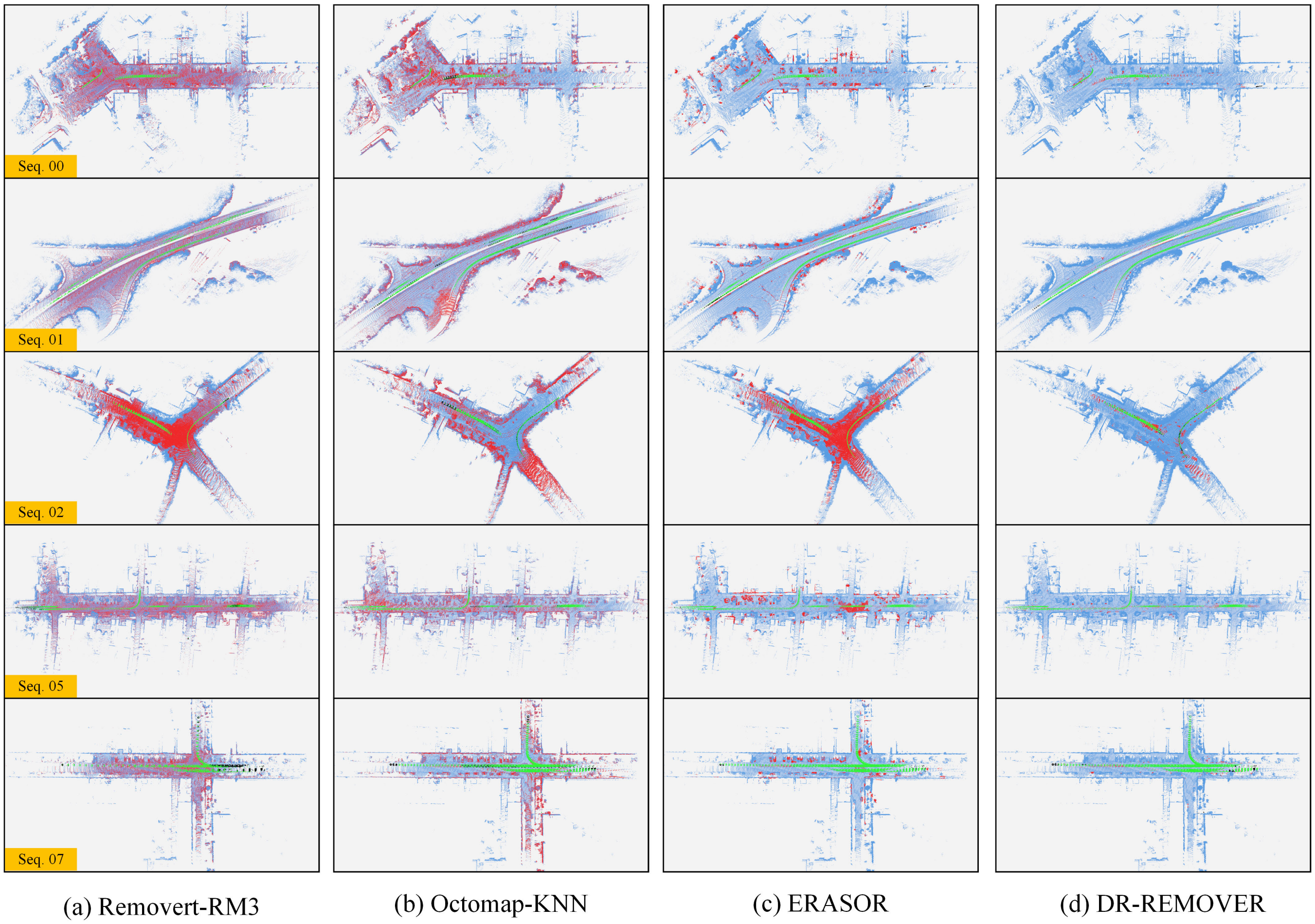


Fig. 6. Comparison of static maps built by DR-REMOVER with other state-of-the-art methods on the SemanticKITTI dataset. The green, red, blue, and black points represent correctly removed dynamic points (TP), falsely removed static points (FP), preserved static points (TN), and preserved dynamic points (FN) on the static point cloud map, respectively. Best viewed in color.

points. After processing each dynamic Group in $\mathcal{P}_t^M$, the local static map $\hat{\mathcal{M}}_t^W$ created is denoted as:

$$\hat{\mathcal{M}}_t = {}_Q^W\mathbf{T}_t \boxplus (\mathcal{P}_t^M - \sum_{i=1}^{N_x}\sum_{j=1}^{N_y} \mathcal{D}_{t,(i,j)}^M), \tag{23}$$

where $\mathcal{D}_{t,(i,j)}^M$ is the set of dynamic points removed from the (i,j)-th Group at time t.

IV. Experimental Evaluation

A. Dataset and Experimental Settings

1) SemanticKITTI dataset. SemanticKITTI's LiDAR scanning data was collected using a Velodyne HDL-64E LiDAR in different rural and urban roads. We selected sequence 00 (4390-4530), sequence 01 (150-250), sequence 02 (860-950), sequence 05 (2350-2670), and sequence 07 (630-820) for our experiments, where the frame number ranges are in parentheses. The classes labeled 252, 253, 254, 255, 256, 257, and 259 were used as dynamic objects to be removed based on SemanticKITTI's labels.
2) Apollo dataset. The Apollo dataset provides LiDAR data from urban scenes in the United States, with a different environmental appearance than SemanticKITTI. The scan data was collected using a Velodyne HDL-64E LiDAR mounted on the roof of the vehicle. We used the results labeled by Arora et al. [15] and selected sequence 00 (0-350), sequence 01 (700-950), sequence 02 (150-450), sequence 03 (0-450), and sequence 04 (900-1550) for our experiments, where the frame number ranges are in parentheses.
3) SemanticPOSS [41] dataset. SemanticPOSS is a publicly available campus dataset collected using Pandora (40-line) LiDAR at Peking University in China. The dataset contains a large number of dynamic instances, especially dominated by pedestrians. Unlike SemanticKITTI and Apollo dataset, the motion trajectories of dynamic objects in SemanticPOSS are cluttered, which makes our method more challenging. We selected Sequence 00 (0-370), Sequence 01 (0-300), Sequence 02 (0-500), and Sequence 05 (0-200) to study the limitations of DR-REMOVER. SemanticPOSS does not have access to the truth values of dynamic objects, and we only use it for qualitative evaluation.

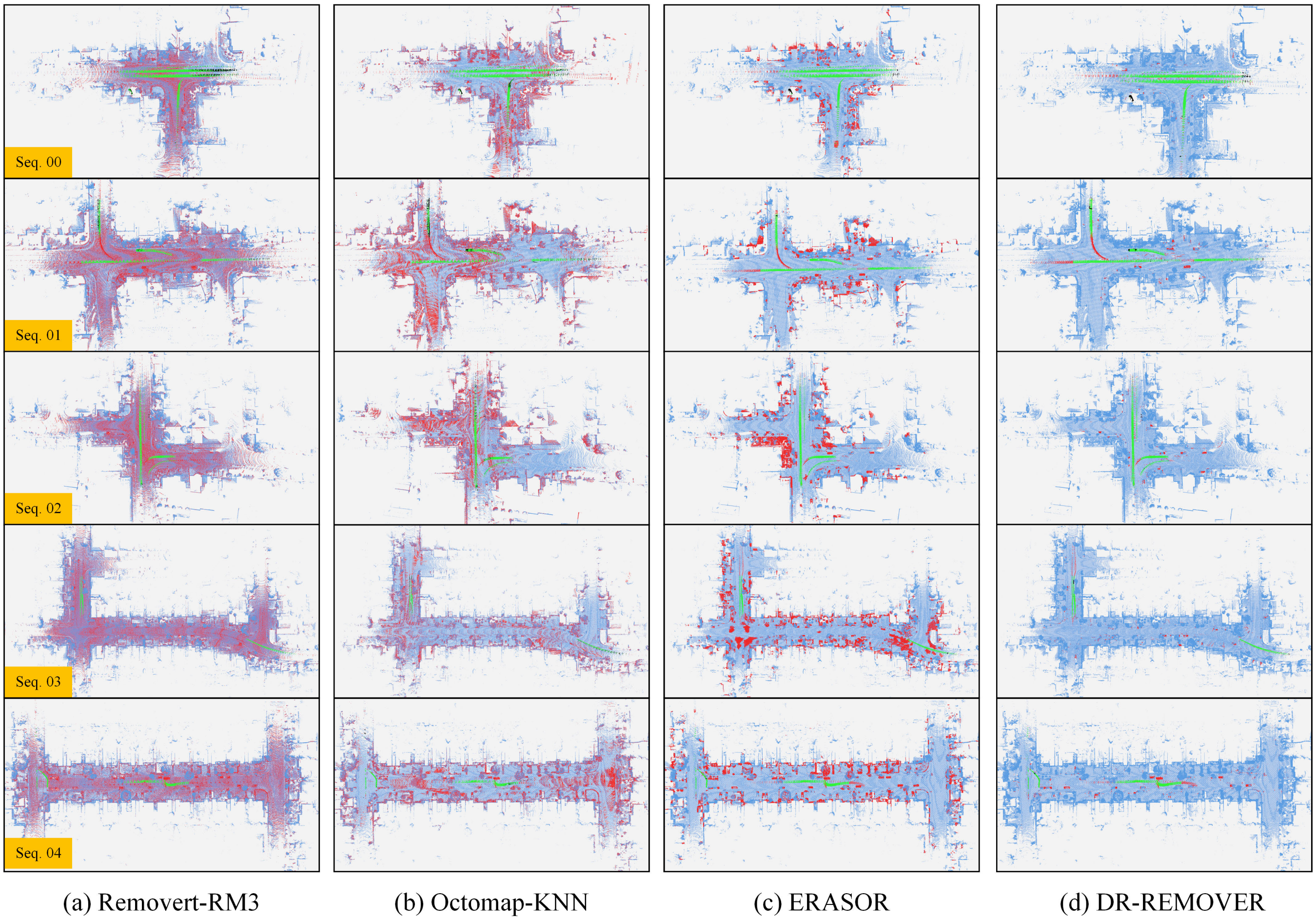


Fig. 7. Comparison of static maps built by DR-REMOVER with other state-of-the-art methods on the Apollo dataset. The green, red, blue and black points represent correctly removed dynamic points (TP), falsely removed static points (FP), preserved static points (TN) and preserved dynamic points (FN) on the static point cloud map, respectively. Maps contain a few inaccurate labels. Best viewed in color.

The static maps built by all methods are quantitatively evaluated on the SemanticKITTI and Apollo datasets. The metrics are calculated voxel-wise, which is the same with [23]. The evaluation metrics PR , RR and F_1 score are defined as follows:

- $PR : \frac{\text{\# of preserved static points}}{\text{\# of total static points on the raw map}}$
- $RR : 1 - \frac{\text{\# of preserved dynamic points}}{\text{\# of total dynamic points on the raw map}}$
- $F_1\ score:\ 2PR \cdot RR/(PR + RR)$

Raw maps are built on SemanticKITTI and Apollo dataset using poses predicted by SuMa [42], which contains inaccuracies in the pose prediction. Raw Maps are built using ground truth on the SemanticPOSS dataset. We use the threshold $thres = 5$ to label dynamic Groups. The parameters used to define FOD are set to $N_s = 120$, $\kappa = 2$ m. We set $h_{seed} = 0.2$ and $\kappa_g = 0.15$ for the ground-plane fitting. All experiments were performed on the same system with an Intel i5-12500H CPU, 2.50GHz, and 16GB RAM.

B. Static Map Building Performance

In this section, DR-REMOVER is comprehensively compared with other state-of-the-art methods. OctoMap-0.05, OctoMap-0.2 and Peopleremover use the data results provided by ERASOR, where 0.05 and 0.2 denote the downsampled voxels used. Octomap-KNN, Removert-RM3, and ERASOR were re-experimented using the source code provided by the authors, and the rest of the methods directly used the results from their papers due to having the same evaluation method. Removert-RM3 indicates the removal of dynamic points using three resolutions of 0.4°, 0.45°, 0.5° respectively, and does not include the reverting session for static points. Octomap-KNN, ERASOR and DR-REMOVER uniformly downsample point clouds using 0.2 m^3 voxels. We use the default downsampling parameter of 0.05 m^3 voxel since Removert is resolution-sensitive. Moreover, DR-REMOVER and ERASOR use the same raw map generation method.

Figs. 6 and 7 visualize the static point cloud maps built by the state-of-the-art method. Obviously, DR-REMOVER builds static point cloud maps with higher quality. As shown in Table I, Octomap and Peopleremover have more serious false positives for static points, since both scan occlusion-based methods do not take into account occlusions caused by static objects. Compared with Octomap, Octomap-KNN uses ground segmentation to separate ground points and builds Octomap using only non-ground points, which significantly improves PR. Both DynamicFilter and Removert are significantly lim-

TABLE I
QUANTITATIVE COMPARISON RESULTS FOR STATIC MAPS ON THE SEMANTICKITTI DATASET. THE BEST SCORES ARE MARKED IN BOLD.

Sequence	Method	PR [%]	RR [%]	F_1 score
00	OctoMap - 0.05 [16]	76.731	99.124	0.865
	OctoMap - 0.2 [16]	34.568	**99.979**	0.514
	OctoMap - KNN [15]	88.238	91.055	0.896
	Peopleremover [20]	37.523	89.116	0.528
	Removert - RM3 [9]	90.228	99.546	0.947
	DynamicFilter [37]	90.070	91.090	0.906
	Auto-MOS [35]	86.164	94.329	0.901
	ERASOR [23]	90.919	98.374	0.945
	Park et al. [5]	94.050	97.190	0.956
	DORF [11]	93.900	98.390	0.961
	DR-REMOVER	**98.117**	98.457	**0.983**
01	OctoMap - 0.05 [16]	53.163	99.660	0.693
	OctoMap - 0.2 [16]	20.777	**99.863**	0.344
	OctoMap - KNN [15]	81.393	77.879	0.796
	Peopleremover [20]	36.349	93.116	0.523
	Removert - RM3 [9]	87.206	93.809	0.904
	DynamicFilter [37]	87.950	87.690	0.878
	Auto-MOS [35]	89.597	90.130	0.899
	ERASOR [23]	91.886	94.802	0.933
	Park et al. [5]	91.815	94.096	0.929
	DORF [11]	97.340	93.970	0.956
	DR-REMOVER	**99.530**	98.335	**0.989**
02	OctoMap - 0.05 [16]	54.112	98.769	0.699
	OctoMap - 0.2 [16]	23.746	**99.792**	0.384
	OctoMap - KNN [15]	84.171	85.587	0.849
	Peopleremover [20]	29.037	94.527	0.444
	Removert - RM3 [9]	74.036	94.975	0.832
	DynamicFilter [37]	88.020	86.100	0.871
	Auto-MOS [35]	88.539	93.352	0.909
	ERASOR [23]	79.461	99.413	0.883
	Park et al. [5]	91.208	95.510	0.933
	DORF [11]	88.680	91.230	0.899
	DR-REMOVER	**97.101**	95.568	**0.963**
05	OctoMap - 0.05 [16]	76.341	96.785	0.854
	OctoMap - 0.2 [16]	33.904	**99.882**	0.506
	OctoMap - KNN [15]	90.058	93.564	0.918
	Peopleremover [20]	38.495	90.631	0.540
	Removert - RM3 [9]	88.758	91.763	0.902
	DynamicFilter [37]	90.170	84.650	0.873
	Auto-MOS [35]	87.931	72.333	0.794
	ERASOR [23]	88.553	98.311	0.932
	Park et al. [5]	93.820	95.740	0.947
	DORF [11]	95.240	92.680	0.939
	DR-REMOVER	**99.328**	98.941	**0.991**
07	OctoMap - 0.05 [16]	77.838	96.938	0.863
	OctoMap - 0.2 [16]	38.183	**99.565**	0.552
	OctoMap - KNN [15]	86.409	90.095	0.882
	Peopleremover [20]	34.772	91.983	0.505
	Removert - RM3 [9]	90.958	86.804	0.888
	DynamicFilter [37]	87.940	86.800	0.874
	Auto-MOS [35]	86.641	91.785	0.891
	ERASOR [23]	90.965	98.807	0.947
	Park et al. [5]	91.146	97.284	0.941
	DORF [11]	92.800	76.650	0.840
	DR-REMOVER	**96.295**	98.524	**0.974**

TABLE II
QUANTITATIVE COMPARISON RESULTS FOR STATIC MAPS ON THE APOLLO DATASET. THE BEST SCORES ARE MARKED IN BOLD.

Sequence	Method	PR [%]	RR [%]	F_1 score
00	OctoMap - KNN [15]	88.208	94.487	0.912
	Removert - RM3 [9]	90.228	**99.546**	0.947
	ERASOR [23]	87.704	94.487	0.912
	DR-REMOVER	**99.132**	97.806	**0.985**
01	OctoMap - KNN [15]	80.962	85.246	0.830
	Removert - RM3 [9]	91.413	91.595	0.915
	ERASOR [23]	87.148	**98.777**	0.926
	DR-REMOVER	**98.608**	96.877	**0.977**
02	OctoMap - KNN [15]	79.358	94.276	0.862
	Removert - RM3 [9]	90.526	84.459	0.874
	ERASOR [23]	83.594	**99.696**	0.909
	DR-REMOVER	**98.909**	99.541	**0.992**
03	OctoMap - KNN [15]	78.944	92.054	0.850
	Removert - RM3 [9]	90.718	**99.128**	0.947
	ERASOR [23]	79.487	98.954	0.882
	DR-REMOVER	**99.244**	98.094	**0.987**
04	OctoMap - KNN [15]	82.268	96.624	0.889
	Removert - RM3 [9]	90.485	**99.460**	0.948
	ERASOR [23]	80.593	99.009	0.889
	DR-REMOVER	**98.550**	99.195	**0.989**

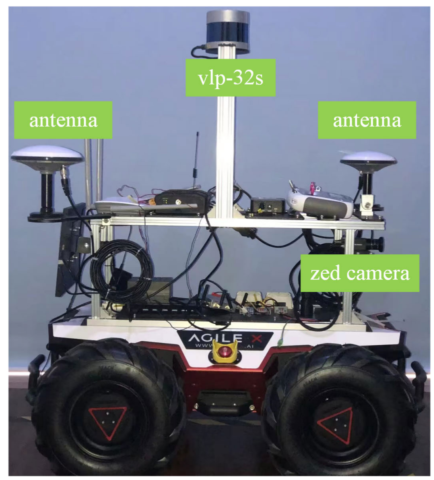


Fig. 8. UGV data collection platform.

ited by incidence angle ambiguity and line-of-sight occlusion, preventing them from exhibiting excellent performance. Although ERASOR significantly increases PR while maintaining a high RR, ERASOR still incorrectly deletes hundreds of thousands of static points on the Apollo dataset. Furthermore, ERASOR relies on efficient ground plane fitting, making it difficult to remove dynamic objects without ground points in SemanticKITT 01.

Benefiting from dynamic Group's accurate recognition and the strategy of reverting static points, DR-REMOVER maintains high PR and RR on all sequences. DR-REMOVER can remove dynamic objects in regions without ground points using the height information of neighboring Groups, even on SemanticKITTI 01 with multiple lanes. However, our method fails to remove dynamic points hidden in the ground layer limited by the accuracy of mapping, resulting in lower RR value on SemanticKITTI 02.

C. Static Map Building in Highly Crowded Environments

We use the campus dataset collected by the unmanned vehicle platform to investigate the performance of DR-REMOVER in a highly crowded environment. Fig. 8 displays the data collection platform, which is equipped with a Velodyne-32C LiDAR, a ZED camera with built-in IMU, and dual antennas. Fig. 9 shows the data collection environment captured by the ZED camera. The road is flat with thick trees on both sides of the road. We choose to collect data by maneuvering the UGV on the roadway for one round trip in the evening when

TABLE III
ABLATION STUDY ON SEMANTICKITTI 02 AND 05 (METRIC VALUE ON SEQ. 02 / METRIC VALUE ON SEQ. 05). COMPONENTS PRESENT IN THE SYSTEM ARE LABELED USING CHECK MARKS.

Full Occlusion Distance (FOD)	Area Occupancy Ratio (AOR)	Reverting Ground Points (Section III.E)	# of FP	PR [%]	RR [%]	F_1 score
			72.56k/72.45k	81.10/92.70	99.45/99.21	89.34/95.95
✓			72.56k/20.61k	81.10/98.19	99.45/99.17	89.34/98.68
✓	✓		26.16k/13.26k	92.77/98.83	98.66/98.62	95.62/98.72
✓	✓	✓	7.83k/6.84k	97.10/99.33	95.57/98.94	96.33/99.13

Fig. 9. UGV data collection environment.

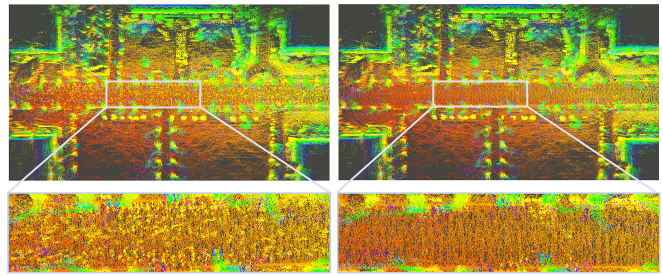


Fig. 10. Comparison of the original map and the static map built by DR-REMOVER.The yellowish clutter of points on the road indicates pedestrians.

pedestrians are at their highest. The original map is built using keyframe poses estimated by the LIO-SAM [43] algorithm.

Fig. 10 visualizes the original map and the static map built by DR-REMOVER. Colors with strong contrast are used to show the effect of dynamic object removal since the real labels are lacking. Although dynamic points in the original map are filled with roads, DR-REMOVER still accurately removes points indicating pedestrians.

Fig. 11 illustrates the static maps built by DR-REMOVER with two other state-of-the-art methods. Unlike the experimental results on urban roads, Removert-RM3 fails in crowded scenarios with complex changes in visibility. ERASOR is affected by noisy points and limited by the ground-plane fitting precision, making the false detection of static points even worse in this scenario. Nevertheless, DR-REMOVER still exhibits excellent performance, demonstrating the effectiveness of using dual-resolution grids for dynamic points removal.

D. Ablation Study

In this section, the validity of each component of DR-REMOVER is analyzed to better understand the proposed method. We studied the impact on the system of three components on SemanticKITTI 02 and 05: FOD, AOR, and Reverting Ground Points. As depicted in the static maps on these two sequences in Fig. 6, Seq. 02 and Seq. 05 have significantly different road environments. Seq. 02 has curved roads with many branches with steep slopes, while Seq. 05 has longer straight roads.

As shown in Table III, the effects of the different components are clearly different on the two sequences. For instance, FOD is ineffective on Seq. 05, while it reverts to most of the static points on Seq. 02. In fact, the ability of each component to revert static points is clearly influenced by the environment and road conditions, and no component plays a fixed dominant role. In other words, all components are utilizing different features to jointly reduce the misdetection of static points, which is what makes DR-REMOVER shows better performance on all sequences tested.

E. Hyperparameter Setting and Comparison

DR-REMOVER's ability to remove dynamic points is well demonstrated in previous experiments. In fact, this ability relies on the accurate detection of dynamic Groups in the early stage. The Group resolution and $thres$ undoubtedly play a decisive role during extracting dynamic Groups. Therefore, we further studied the effect of Group resolution and $thres$ on DR-REMOVER. This experiment did not use the system components used for reverting static points.

As shown in Fig. 12(a), the proposed method does not require the precise value of $thres$ since DR-REMOVER maintains a good performance within an interval of values. This also demonstrates the validity of occupied heights being used to extract dynamic regions. As shown in Fig. 12(b), our method maintains better performance when the Group resolution is in the range of 2 m to 4 m. DR-REMOVER hardly recognizes dynamic regions when the Group resolution is 0.5 m. This result validates the previous clarification that higher resolution grids are not conducive to capturing the overall characteristics of dynamic objects. Groups with higher resolutions are more significantly affected by point cloud sparsity, and Groups with lower resolutions tend to contain both dynamic and static objects.

F. Runtime Evaluation

Moreover, we compare the runtimes of the different methods. The time for all methods to process each frame of scan data is significantly affected by the number of dynamic points,

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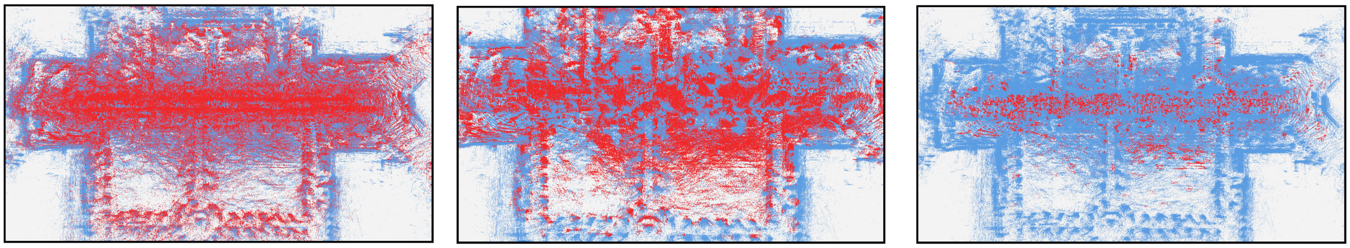
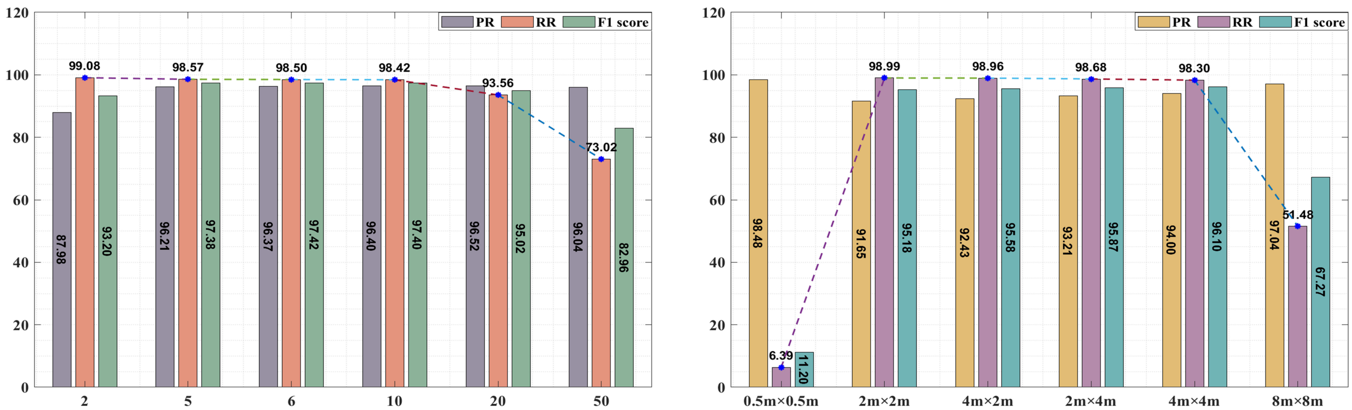


(a) Removert-RM3 (b) ERASOR (c) DR-REMOVER

Fig. 11. Static maps built by DR-REMOVER in highly crowded environments. Red points indicate removal from the original map. Best viewed in color.



(a) *thres* (b) Group resolution

Fig. 12. Hyperparametric experiments for and Group resolution on SemanticKITTI 07.

TABLE IV
AVERAGE RUNTIME OF DR-REMOVER AND OTHER STATE-OF-THE-ART METHODS ON SEMANTICKITTI 01.

Method	Preprocessing time(s)	Removal time(s)	Total time (s)
Octomap-KNN	4.250	4.510	8.760
Removert-RM3	0.601	0.030	0.631
Removert-RM3-Mul	0.207	0.036	0.243
ERASOR	0.063	0.041	0.104
DR-REMOVER	0.058	0.036	0.094

so the average running time was used for a more reasonable comparison. Table IV shows the runtime for all methods on SemanticKITTI 01. Sequence 01 is a large scene with an open line of sight and contains more point clouds in each frame of data. Importantly, we take the step of generating a multilayer height map in Octomap-KNN and the step of generating a distance image in Removert-RM3 as their preprocessing processes, respectively. Removert-RM3-Mul denotes the use of multithreading of the source code. The processes of ERASOR and DR-REMOVER to generate descriptors are considered as preprocessing. Although DR-REMOVER has more steps for dynamic point removal, dynamic Groups do not necessarily perform all of these steps. Compared with other state-of-the-art methods, DR-REMOVER takes less time to run.

As shown in Table IV, Removert, ERASOR, and DR-REMOVER have similar removal times, while the preprocessing process for Removert is significantly more time-consuming. On the one hand, Removet does not utilize height values to filter the point cloud, thus requiring more points to be processed. On the other hand, Removet creates and maintains distance images with high resolution. Octomap-KNN, as a ray-tracing based method, has the most time-consuming preprocessing and dynamic point removal processes. Octomap-KNN not only needs to maintain high-resolution voxel maps and update the occupancy state of each voxel in the map, but also needs to perform canny edge detection on multilayer height maps to extract ground points.

G. Limitation

The SemanticPOSS dataset with complex dynamic instances is used to explore DR-REMOVER failure scenarios. As shown in Fig. 13, the roads of the original map are filled with large moving crowds, and the narrow roads are lined with lush trees and parked cars. DR-REMOVER removes most of the dynamic objects on the road without losing many static points. However, many dynamic objects remain in the original map that have not been removed. Our method is limited for dynamic points rejection in these challenging scenarios. On the one hand, the process of extracting dynamic Groups is sensitive to height values, and our method has difficulty removing dynamic objects hidden under trees with lower heights. On the other hand, the proposed method has difficulty in recognizing dynamic objects that are immediately adjacent to static objects. This Group is incorrectly determined to be

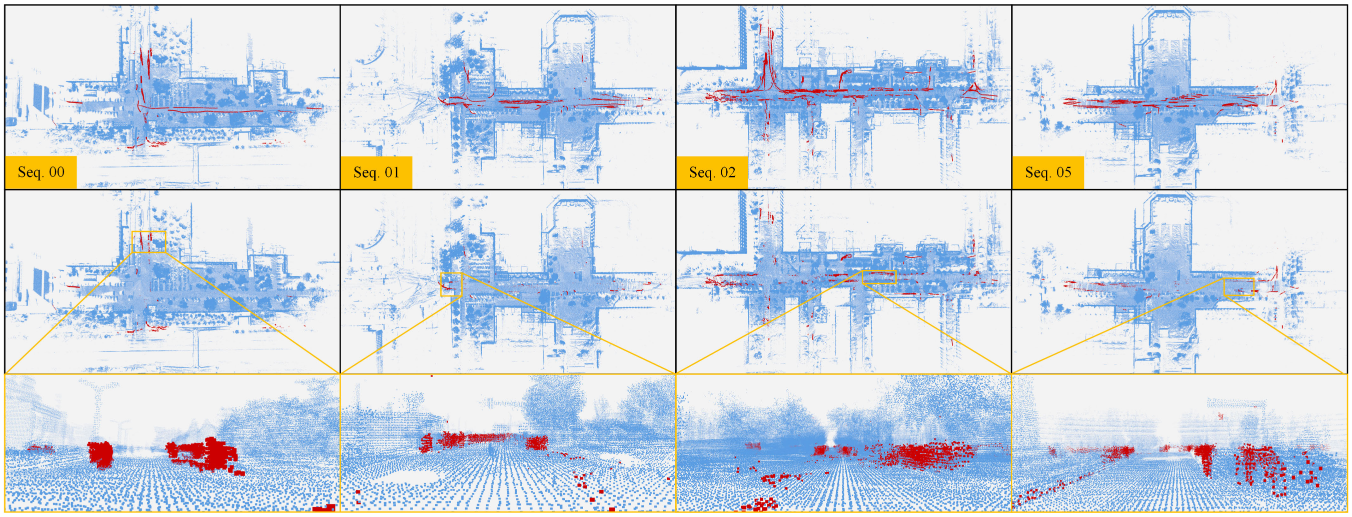


Fig. 13. Static maps built by DR-REMOVER in highly crowded environments. The figure in the first row represents the original map and the second row represents the static point cloud map created after DR-REMOVER removes the dynamic points. The red points indicate real dynamic points. Maps contain a few inaccurate labels. Best viewed in color.

static by DR-REMOVER when dynamic and static objects are divided into a Group .

V. Conclusion

In this letter, the method called DR-REMOVER is proposed for removing dynamic objects to build highly accurate static maps. Our method combines the advantages of two grids with different resolutions to achieve accurate detection of dynamic points. As demonstrated by experiments on the SemanticKITTI and Apollo datasets, DR-REMOVER exhibits static points preservation capabilities that other state-of-the-art methods do not have. In addition, DR-REMOVER has good real-time performance.

In future work, we will study the robustness of grid-based methods to different grid resolutions. Furthermore, we will focus on solving the dynamic points removal problem in highly crowded environments and improving DR-REMOVER into an online static map building method.

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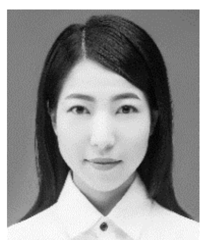

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